off & on again

Intrusion Report #1 - Team 3

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Overview

Multiple RDP brute-force attempts were detected on our Windows servers. In response, Fail2Ban successfully identified and blocked the offending IP addresses. During this period, the logs show a high volume of session activity, including repeated login attempts, authentication failures, and connections initiated under multiple user accounts. Concurrently, legitimate database activity from warehouse personnel continued as expected.

AD

Failed login spam attempts

At 8:06AM, multiple failed login attempts were captured by Event Viewer. From the security details, a computer named `kali` failed to log on using multiple default passwords, default accounts, and passwordless login. Due to our strict security settings, the threat actor was not allowed access. We further confirmed that this user was not connected afterwards.

Event 4625, Microsoft Windows security auditing.

General | Details

```
Logon Type:                     3

Account For Which Logon Failed:
        Security ID:            NULL SID
        Account Name:           user
        Account Domain:         FAKE

Failure Information:
        Failure Reason:         Unknown user name or bad password.
        Status:                 0xC000006D
```

Log Name:	Security		
Source:	Microsoft Windows security	Logged:	4/25/2026 8:06:27 AM
Event ID:	4625	Task Category:	Logon
Level:	Information	Keywords:	Audit Failure
User:	N/A	Computer:	LP.team3.isucdc.com
OpCode:	Info		
More Information:	Event Log Online Help		

```
Account For Which Logon Failed:
        Security ID:            NULL SID
        Account Name:           kali
        Account Domain:         WORKGROUP

Failure Information:
        Failure Reason:         Unknown user name or bad password.
        Status:                 0xC000006D
        Sub Status:             0xC0000064
```

At 08:17 there were multiple RDP attempts on AD and WMS from 12.119.206.199.

Traffic Sniffer

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Time	data.src_ip	data.app_proto	data.fileinfo.filename	data.event_type	data.http.hostname	data.dest_ip
Apr 25, 2026 @ 13:12:09.943	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.937	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.930	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.926	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.919	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.909	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.900	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.898	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:07.920	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:07.913	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:07.907	201.203.200.10	-	-	rdp	-	12.110.206.199

WWW

Failed login attempts - WWW:
At 8:03 AM, we had multiple failed password checks/authentication failures on WWW. We identified this initially from our Wazuh logging platform and tailed the in-depth var/log/secure to watch for login attempts. By inspecting these logs, we were able to confirm that our fail2ban configuration and SSHD authentication restrictions successfully mitigated brute force attempts from these actors.

```
Apr 25 07:59:58 www sshd[371686]: pam_unix(sshd:session): session closed for user jacqueline.elliott
Apr 25 08:00:08 www systemd[371696]: pam_unix(systemd-user:session): session closed for user jacqueline.el
liott
Apr 25 08:01:49 www unix_chkpwd[372498]: password check failed for user (cdc)
Apr 25 08:01:49 www sshd[372496]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=s
sh ruser= rhost=12.110.242.119  user=cdc
Apr 25 08:01:49 www sshd[372496]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ss
h ruser= rhost=12.110.242.119 user=cdc
Apr 25 08:01:49 www sshd[372496]: pam_sss(sshd:auth): received for user cdc: 10 (User not known to the und
erlying authentication module)
Apr 25 08:02:59 www unix_chkpwd[373263]: password check failed for user (cdc)
Apr 25 08:02:59 www sshd[373261]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=s
sh ruser= rhost=12.110.242.123  user=cdc
Apr 25 08:02:59 www sshd[373261]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ss
h ruser= rhost=12.110.242.123 user=cdc
Apr 25 08:02:59 www sshd[373261]: pam_sss(sshd:auth): received for user cdc: 10 (User not known to the und
erlying authentication module)
```

The user ‘cdc’ was checked by a threat actor. This request was denied because only IT Administrators may SSH into this machine.

```
Apr 25 08:05:08 www systemd[1]: Stopped User runtime directory /run/user/443201181.
Apr 25 08:05:09 www systemd[1]: Removed slice User Slice of UID 443201181.
Apr 25 08:06:21 www sshd[374834]: Connection from 12.110.242.118 port 56130 on 192.168.1.20 port 22
Apr 25 08:06:21 www sshd[374836]: Connection from 12.110.242.118 port 56134 on 192.168.1.20 port 22
Apr 25 08:06:21 www sshd[374836]: User cdc from 12.110.242.118 not allowed because none of user's groups a
re listed in AllowGroups
Apr 25 08:06:21 www fail2ban.filter[956]: INFO [sshd] Found 12.110.242.118 - 2026-04-25 08:06:21
Apr 25 08:06:23 www sshd[374836]: Failed password for invalid user cdc from 12.110.242.118 port 56134 ssh2
Apr 25 08:06:24 www fail2ban.filter[956]: INFO [sshd] Found 12.110.242.118 - 2026-04-25 08:06:23
Apr 25 08:06:25 www sshd[374834]: Connection closed by 12.110.242.118 port 56130 [preauth]
Apr 25 08:06:25 www sshd[374836]: Connection closed by invalid user cdc 12.110.242.118 port 56134 [preauth
]

[root@www ~]#
```

Over the span of 08:08 through 08:45 there were several login attempts by multiple users when checking secure logs on WWW

```
]
Apr 25 08:08:49 www sshd[375616]: Connection from 12.110.206.199 port 53792 on 192.168.1.20 port 22
Apr 25 08:08:49 www sshd[375618]: Connection from 12.110.206.199 port 53794 on 192.168.1.20 port 22
Apr 25 08:08:49 www sshd[375618]: User cdc from 12.110.206.199 not allowed because none of user's groups a
re listed in AllowGroups
Apr 25 08:08:50 www fail2ban.filter[956]: INFO [sshd] Found 12.110.206.199 - 2026-04-25 08:08:49
Apr 25 08:08:50 www fail2ban.actions[956]: NOTICE [sshd] Ban 12.110.206.199
Apr 25 08:08:51 www sshd[375618]: Failed password for invalid user cdc from 12.110.206.199 port 53794 ssh2
Apr 25 08:08:51 www fail2ban.filter[956]: INFO [sshd] Found 12.110.206.199 - 2026-04-25 08:08:51
```

```
=0 tty=ssh ruser= rhost=12.110.242.119 user=cdc
Apr 25 08:45:45 www sshd[392233]: pam_sss(sshd:auth): received for user cdc: 10 (User not known t
o the underlying authentication module)
Apr 25 08:45:45 www unix_chkpwd[392243]: password check failed for user (cdc)
Apr 25 08:45:45 www sshd[392223]: pam_unix(sshd:auth): authentication failure; logname= uid=0 eui
d=0 tty=ssh ruser= rhost=12.110.242.119  user=cdc
Apr 25 08:45:45 www sshd[392223]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid
=0 tty=ssh ruser= rhost=12.110.242.119 user=cdc
Apr 25 08:45:45 www sshd[392223]: pam_sss(sshd:auth): received for user cdc: 10 (User not known t
o the underlying authentication module)
Apr 25 08:45:46 www unix_chkpwd[392268]: password check failed for user (cdc)
Apr 25 08:45:46 www sshd[392242]: pam_unix(sshd:auth): authentication failure; logname= uid=0 eui
d=0 tty=ssh ruser= rhost=12.110.242.119  user=cdc
Apr 25 08:45:46 www sshd[392242]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid
=0 tty=ssh ruser= rhost=12.110.242.119 user=cdc
Apr 25 08:45:46 www sshd[392242]: pam_sss(sshd:auth): received for user cdc: 10 (User not known t
o the underlying authentication module)
```

LP

Along with SMB access,

Apr 25, 2026 @ 14:41:28.549	LP	christy.davis	49.10.235.223	User: TEAM3\christy.davis logged using Remote Desktop Connection (RDP) from ip:49.10.235.223.
Apr 25, 2026 @ 14:41:28.525	LP	christy.davis	49.10.235.223	User: TEAM3\christy.davis logged using Remote Desktop Connection (RDP) from ip:49.10.235.223.
Apr 25, 2026 @ 14:38:13.736	ad	christy.davis	49.10.235.223	User: TEAM3\christy.davis logged using Remote Desktop Connection (RDP) from ip:49.10.235.223.
Apr 25, 2026 @ 14:38:13.722	ad	christy.davis	49.10.235.223	User: TEAM3\christy.davis logged using Remote Desktop Connection (RDP) from ip:49.10.235.223.

WMS

At 08:17 there were multiple RDP attempts on WMS and AD from 12.119.206.199.

Traffic Sniffer

1–48 of 48 < 1 >

Time	data.src_ip	data.app_proto	data.fileinfo.filename	data.event_type	data.http.hostname	data.dest_ip
Apr 25, 2026 @ 13:12:09.943	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.937	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.930	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.926	201.203.200.30	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.919	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.909	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.900	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:09.898	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:07.920	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:07.913	201.203.200.10	-	-	rdp	-	12.110.206.199
Apr 25, 2026 @ 13:12:07.907	201.203.200.10	-	-	rdp	-	12.110.206.199

At 08:17 there was a sshd ban success on WMS by fail2ban for 12.119.206.199.

```
Apr 25 13:03:33 wms.team3.isucdc.com fail2ban.filter[731]: INFO [sshd] Found 12.110.206.199 - 2026-04-25 13:03:33
Apr 25 13:03:36 wms.team3.isucdc.com fail2ban.filter[731]: INFO [sshd] Found 12.110.206.199 - 2026-04-25 13:03:36
Apr 25 13:06:36 wms.team3.isucdc.com fail2ban.filter[731]: INFO [sshd] Found 12.110.242.118 - 2026-04-25 13:06:36
Apr 25 13:06:38 wms.team3.isucdc.com fail2ban.filter[731]: INFO [sshd] Found 12.110.242.118 - 2026-04-25 13:06:38
Apr 25 13:08:55 wms.team3.isucdc.com fail2ban.filter[731]: INFO [sshd] Found 12.110.206.199 - 2026-04-25 13:08:55
Apr 25 13:08:55 wms.team3.isucdc.com fail2ban.actions[731]: NOTICE [sshd] Ban 12.110.206.199
Apr 25 13:08:58 wms.team3.isucdc.com fail2ban.filter[731]: INFO [sshd] Found 12.110.206.199 - 2026-04-25 13:08:57
```

DB

Access denied for Richard SQL Logs - DB:
At 8:05 AM, there was a connection attempt on MySQL for the default user, Richard. We identified this by monitoring mysql.log to check suspicious activity. By changing the credentials of this box and allowing only Warehouse Workers to authenticate, we successfully mitigated a default credential access check.

```
host-03 - Proxmox Console - Google Chrome
pve.iseage.org/?console=kvm&xtermjs=1&vmid=174&vmname=Team-03-DB&node=host-03&cmd=
_names, @@max_allowed_packet AS max_allowed_packet, @@net_write_timeout AS net_write_timeout, @@performan
ce_schema AS performance_schema, @@query_cache_size AS query_cache_size, @@query_cache_type AS query_cache
_type, @@sql_mode AS sql_mode, @@system_time_zone AS system_time_zone, @@time_zone AS time_zone, @@transac
tion_isolation AS transaction_isolation, @@wait_timeout AS wait_timeout
2026-04-25T13:03:26.418600Z       372 Query     SET character_set_results = NULL
2026-04-25T13:03:26.445153Z       372 Query     SET autocommit=1
2026-04-25T13:03:26.446349Z       372 Query     SELECT @@session.transaction_read_only
2026-04-25T13:03:57.337141Z       374 Connect   richard@12.110.206.199 on  using TCP/IP
2026-04-25T13:03:57.337212Z       374 Connect   Access denied for user 'richard'@'12.110.206.199' (using p
assword: YES)
2026-04-25T13:04:00.930371Z       376 Connect   cdc@12.110.206.199 on  using TCP/IP
2026-04-25T13:04:00.932349Z       376 Connect   Access denied for user 'cdc'@'12.110.206.199' (using passw
ord: YES)
2026-04-25T13:04:28.615213Z       356 Quit
2026-04-25T13:04:28.618280Z       377 Connect   wms_svc@192.168.1.40 on iseage_db using TCP/IP
2026-04-25T13:04:28.620668Z       377 Query     /* mysql-connector-j-8.0.33 (Revision: 7d6b0800528b6b25c68
52dc10d6c1c8429c100c) */SELECT  @@session.auto_increment_increment AS auto_increment_increment, @@charact
er_set_client AS character_set_client, @@character_set_connection AS character_set_connection, @@character
```

At 08:06 there were fail2ban activity logs on Wazuh, two IPs banned.

Apr 25, 2026 @ 13:06	DB	-	-		Fail2ban activity: for IP in jail	Apr 25 13:06: 44 db.team3.isucdc.com fail2ban.actions [1414]: NOTICE [sshd] 12.110.242.118 already banned	-	-	
Apr 25, 2026 @ 13:06:35.005	DB	-	-		Fail2ban activity: for IP in jail	Apr 25 08:06: 44 db fail2ban.actions[1414]: NOTICE [sshd] 12.110.242.118 already banned	-	-	
Apr 25, 2026 @ 14:48:10.119	DB	-	199.100.199.192	Fail2ban activity: Found for IP 199.100.199.192 in jail sshd	Apr 25 09:48: 20 db fail2ban.filter[1414]: INFO [sshd] Found 199.100.199.192	Found	-	-	
Apr 25, 2026 @ 14:48:10.119	DB	-	199.100.199.192	Fail2ban activity: Found for IP 199.100.199.192 in jail sshd	Apr 25 09:48: 20 db fail2ban.filter[1414]: INFO [sshd] Found 199.100.199.192	Found	-	-	

At 08:13 there was a failed login attempt on DB by Richard.

```
pve.iseage.org/?console=kvm&xtermjs=1&vmid=174&vmname=Team-03-DB&node=host-03&cmd=
er_set_client AS character_set_client, @@character_set_connection AS character_set_connection, @@character
_set_results AS character_set_results, @@character_set_server AS character_set_server, @@collation_server
AS collation_server, @@collation_connection AS collation_connection, @@init_connect AS init_connect, @@int
eractive_timeout AS interactive_timeout, @@license AS license, @@lower_case_table_names AS lower_case_tabl
e_names, @@max_allowed_packet AS max_allowed_packet, @@net_write_timeout AS net_write_timeout, @@performan
ce_schema AS performance_schema, @@query_cache_size AS query_cache_size, @@query_cache_type AS query_cache
_type, @@sql_mode AS sql_mode, @@system_time_zone AS system_time_zone, @@time_zone AS time_zone, @@transac
tion_isolation AS transaction_isolation, @@wait_timeout AS wait_timeout
2026-04-25T13:11:59.283852Z       385 Query     SET character_set_results = NULL
2026-04-25T13:11:59.284551Z       385 Query     SET autocommit=1
2026-04-25T13:11:59.285081Z       385 Query     SELECT @@session.transaction_read_only
2026-04-25T13:12:39.357451Z       387 Connect   richard@12.110.206.199 on  using TCP/IP
2026-04-25T13:12:39.359597Z       387 Connect   Access denied for user 'richard'@'12.110.206.199' (using p
assword: YES)
2026-04-25T13:12:39.898903Z       389 Connect   cdc@12.110.206.199 on  using TCP/IP
2026-04-25T13:12:39.901396Z       389 Connect   Access denied for user 'cdc'@'12.110.206.199' (using passw
ord: YES)
```

Multiple sessions open and closed on DB by jennifer.

```
host-03 - Proxmox Console - Google Chrome
pve.iseage.org/?console=kvm&xtermjs=1&vmid=197&vmname=Team-03-WWW&node=host-03&cmd=
n
Apr 25 08:19:57 www sshd[380892]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid
=0 tty=ssh ruser= rhost=12.110.242.125 user=jennifer.freeman
Apr 25 08:19:58 www systemd[380926]: pam_unix(systemd-user:session): session opened for user jenn
ifer.freeman by (uid=0)
Apr 25 08:19:58 www sshd[380892]: pam_unix(sshd:session): session opened for user jennifer.freema
n by (uid=0)
Apr 25 08:19:58 www sshd[380892]: pam_unix(sshd:session): session closed for user jennifer.freema
n
Apr 25 08:24:58 www sshd[382893]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid
=0 tty=ssh ruser= rhost=68.32.254.50 user=jacqueline.elliott
Apr 25 08:24:58 www systemd[382934]: pam_unix(systemd-user:session): session opened for user jacq
ueline.elliott by (uid=0)
Apr 25 08:24:58 www sshd[382893]: pam_unix(sshd:session): session opened for user jacqueline.elli
ott by (uid=0)
Apr 25 08:24:58 www sshd[382893]: pam_unix(sshd:session): session closed for user jacqueline.elli
ott
```

At 08:29 viewed sql.log, determined normal warehouse behavior from a worker.

```
pve.iseage.org/?console=kvm&xtermjs=1&vmid=174&vmname=Team-03-DB&node=host-03&cmd=
2026-04-25T13:28:14.516060Z       400 Connect   anna.pugh@12.110.242.121 on iseage_db using TCP/I
P
2026-04-25T13:28:14.521252Z       400 Query     SELECT VERSION() as v
2026-04-25T13:28:14.581303Z       400 Query     show databases
2026-04-25T13:28:14.604134Z       400 Query     SELECT table_name as name
      FROM information_schema.tables
      WHERE table_schema = database()
      AND table_type NOT LIKE '%VIEW%'
      ORDER BY table_name
2026-04-25T13:28:14.642921Z       400 Query     SELECT table_name as name
      FROM information_schema.views
      WHERE table_schema = database()
      ORDER BY table_name
2026-04-25T13:28:14.657165Z       400 Query     select
        r.specific_name as specific_name,
        r.routine_name as routine_name,
        r.routine_type as routine_type,
        r.data_type as data_type,
```

At 08:59 there was a failed login attempt on DB from sql.log.

```
host-03 - Proxmox Console - Google Chrome
pve.iseage.org/?console=kvm&xtermjs=1&vmid=174&vmname=Team-03-DB&node=host-03&cmd=
pr 25 08:55:00 db systemd-logind[1376]: Removed session 147.
pr 25 08:55:00 db systemd: pam_unix(systemd-user:session): session closed for user christy.davis
C
root@db:~# tail -f /var/log/mysql/mysql.log
2026-04-25T13:54:59.723709Z       410 Query     set session transaction read only
2026-04-25T13:54:59.724365Z       410 Query     SET autocommit=0
2026-04-25T13:54:59.730340Z       410 Query     select i1_0.id,i1_0.description,i1_0.name,i1_0.pr
ice,i1_0.quantity from items i1_0
2026-04-25T13:54:59.733808Z       410 Query     commit
2026-04-25T13:54:59.734321Z       410 Query     SET autocommit=1
2026-04-25T13:54:59.734936Z       410 Query     set session transaction read write
2026-04-25T13:56:11.124053Z       425 Connect   richard@12.110.206.199 on  using TCP/IP
2026-04-25T13:56:11.124160Z       425 Connect   Access denied for user 'richard'@'12.110.206.199'
 (using password: YES)
2026-04-25T13:56:14.718193Z       427 Connect   cdc@12.110.206.199 on  using TCP/IP
2026-04-25T13:56:14.718227Z       427 Connect   Access denied for user 'cdc'@'12.110.206.199' (us
ing password: YES)
```